Horizontal Directional Drilling · 3D explainer storyboard

Directional Technologies, Inc. (DTI) · 16:9, 1920 × 1080 · 3:08 · synchronized to the original recorded narration

Draft for review, 2026-11-03. Frames are rendered from the actual 3D production files (draft quality).

Production requirements and open items

- **Pickup line:** “over the past 23 years” is removed from the original audio. The replacement “for over 30 years.” (highlighted below) is **pending**: a consented voice match of the original narrator requires the narrator's written consent and an approved ElevenLabs credit estimate. Until then the audio is silent there and drafts show a PICKUP PENDING flag.
- **Personas document:** “DTI Personas – 7-15-26.pdf” was not received. Audience focus follows the six audience priorities in the brief; it will be refined when the PDF arrives.
- **Transcript to confirm:** "or radio detection device" (heard as "our", 0.84 confidence): confirm. "one and a half size the conduit" is what the narrator says; captions keep it verbatim: confirm.
- **HDD specialist review** of bore geometry, equipment behavior, tracking graphics, fluid handling and pullback connections is required before the final render (see notes per shot).
- **Music:** none licensed yet. The draft has narration only; a licensed bed will be added at low level under the voice.
- **Contact details** on the end card as supplied: 203-294-9200 · 877-788-4479 · drilling@directionaltech.com · directionaltech.com.
- Utility colors follow the APWA / Common Ground Alliance uniform color code; utility depths, sizes and the site are illustrative, not survey data. No project measurements, costs, schedules or outcome guarantees are shown.



S01 · 0:00.0 to 0:04.1

S01 · Horizontal Directional Drilling

NARRATION (ORIGINAL RECORDING)

No narration (silent opening, as in the original).

VISUAL

High aerial of a contemporary intersection: four lanes each way, signals cycling, steady traffic, ground-floor businesses. Slow push in. No narration yet (original has a silent intro).

ON-SCREEN TEXT

- Horizontal Directional Drilling @ 0:00.8

AUDIENCE FOCUS

Owners and end clients



S02 · 0:04.1 to 0:10.4

S02

NARRATION (ORIGINAL RECORDING)

0:04.1 Directional drilling is used to install conduits under obstacles such as roads, buildings and wetlands.

VISUAL

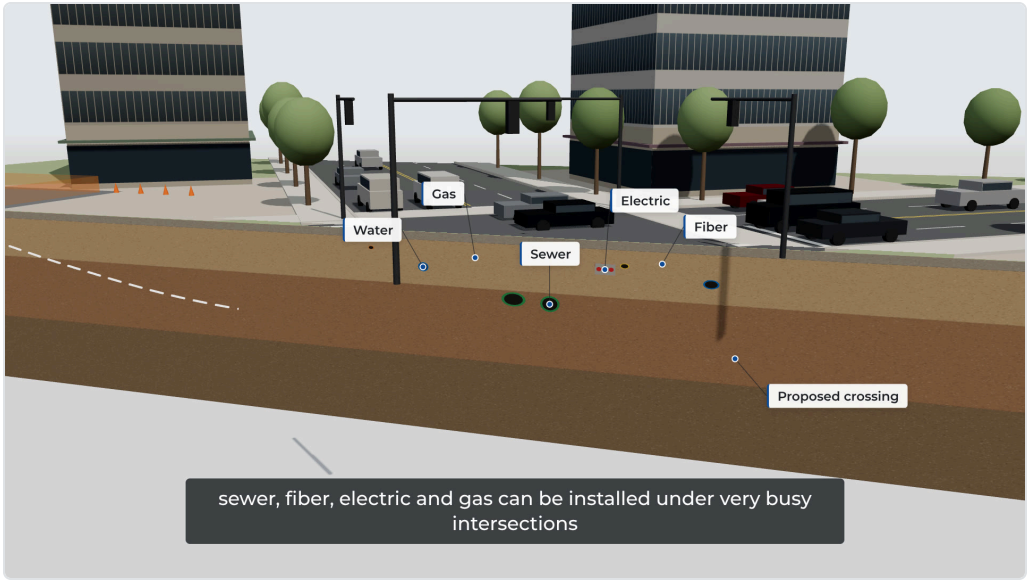
Camera descends to street level along Main St. Short labels for roads and buildings as they are named.

ON-SCREEN TEXT

- Roads @ 0:07.9
- Buildings @ 0:08.5

AUDIENCE FOCUS

Owners and end clients · Civil engineers



S03 · 0:10.4 to 0:22.0

S03

NARRATION (ORIGINAL RECORDING)

0:10.5 As demonstrated in this video, conduits such as water, sewer, fiber, electric and gas can be installed under very busy intersections without disruption of traffic.

VISUAL

The ground opens along the crossing (cutaway between the two pits). Existing utilities appear in section, labeled as named: water, sewer, fiber, electric, gas (APWA colors). The proposed crossing is drawn in white below them. Camera rises so traffic above stays in frame.

ON-SCREEN TEXT

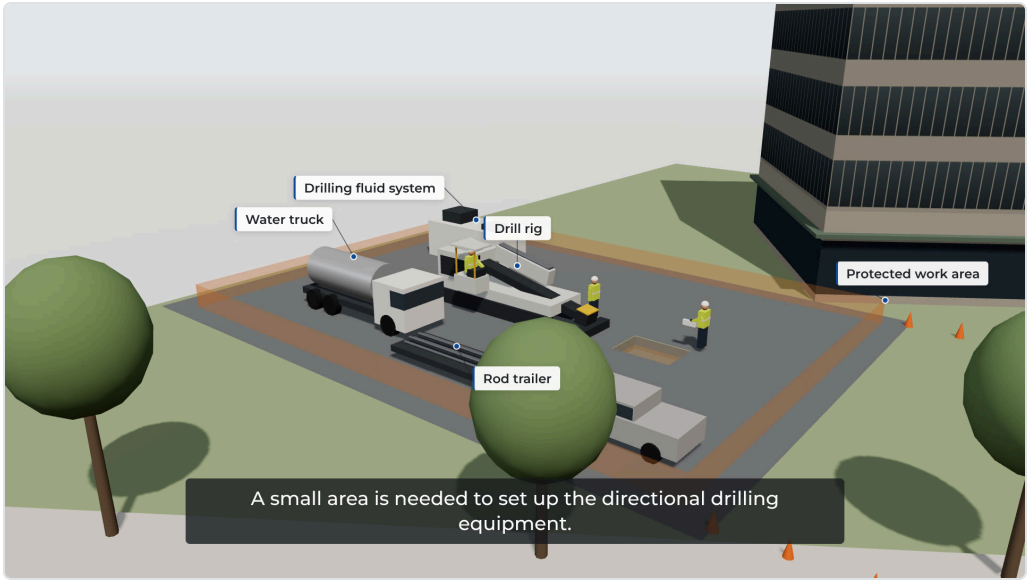
- Water @ 0:13.2
- Sewer @ 0:13.8
- Fiber @ 0:14.4
- Electric @ 0:14.9
- Gas @ 0:15.9
- Proposed crossing @ 0:16.6

AUDIENCE FOCUS

Civil engineers · Estimators · Owners and end clients

HDD SPECIALIST REVIEW

Utility depths and sizes are illustrative. Confirm the bore depth below the deepest utility reads as plausible.



S04 · 0:22.0 to 0:31.6

S04

NARRATION (ORIGINAL RECORDING)

0:22.1 A small area is needed to set up the directional drilling equipment.

0:26.0 This area can be strategically placed away from congestion and business activities.

VISUAL

Surface closes. The drilling setup on the corner lot: rig, drilling fluid system, water truck, rod trailer, crew truck, fenced work area. Then the camera rises to show the lot set back from the traffic lanes and business entrances.

ON-SCREEN TEXT

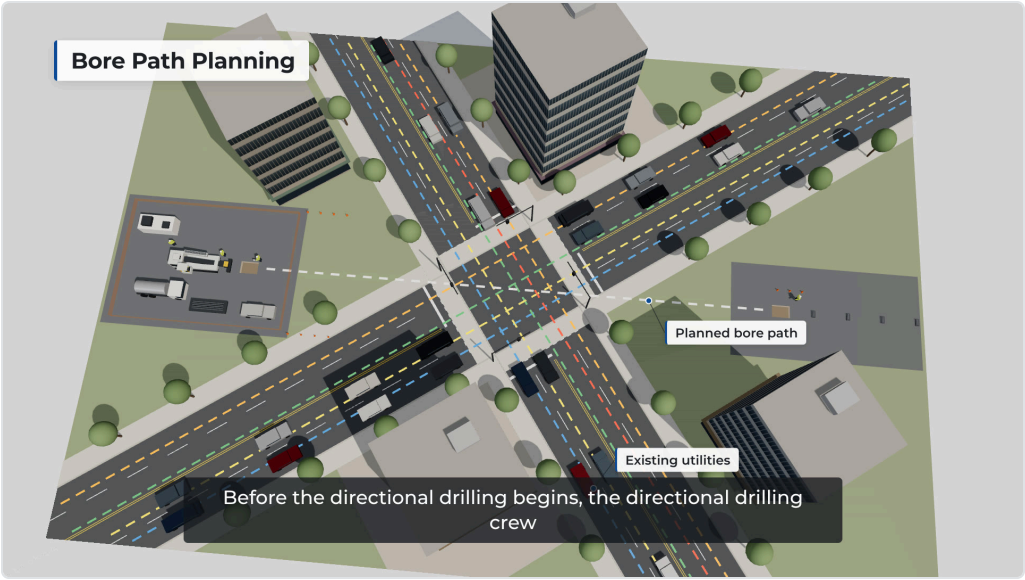
- Drill rig @ 0:22.9
- Drilling fluid system @ 0:23.3
- Water truck @ 0:23.7
- Rod trailer @ 0:24.1
- Protected work area @ 0:24.5
- Work area set back from traffic and business access (outline)

AUDIENCE FOCUS

Project managers · Civil contractors · Owners and end clients

HDD SPECIALIST REVIEW

Equipment layout and footprint: confirm a plausible mid-size rig spread.



S05 · 0:31.6 to 0:48.1

S05 · Bore Path Planning

NARRATION (ORIGINAL RECORDING)

0:31.7 Before the directional drilling begins, the directional drilling crew pre-plans the bore path to carefully avoid obstructions such as other underground utilities.

VISUAL

Plan view with the ground made see-through: existing utilities along both streets; the planned bore path drawn as a white dashed line from the entrance pit to the exit pit (white = proposed excavation, APWA). Surface marks follow the route. Then a profile cutaway shows the planned path passing below the utilities with clearance. Camera moves to the entrance pit.

ON-SCREEN TEXT

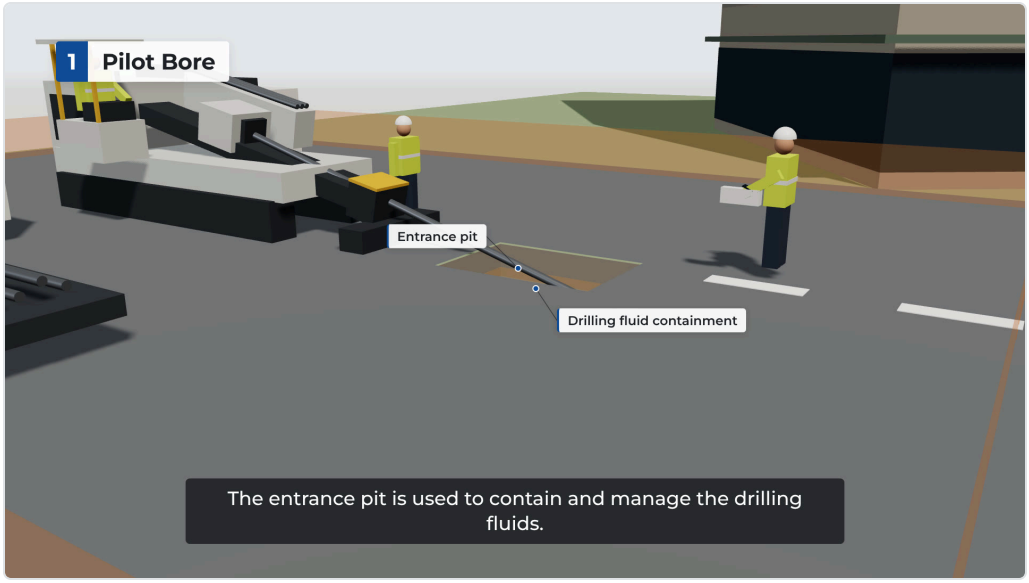
- Bore Path Planning @ 0:31.8
- Existing utilities @ 0:34.0
- Planned bore path @ 0:35.0
- Entrance pit @ 0:35.6
- Exit pit @ 0:35.9
- Clearance below existing utilities (bracket, no values)
- Utility locate marks on the surface (APWA colors)

AUDIENCE FOCUS

Estimators · Civil engineers · Project managers

HDD SPECIALIST REVIEW

Profile shape (entry angle, depth below utilities, bend radius) and the clearance graphic. No measurements are shown.



S06 · 0:48.1 to 1:04.4

S06 · 1. Pilot Bore

NARRATION (ORIGINAL RECORDING)

0:48.2 The horizontal bore begins with the directional drill bit entering the entrance pit.

0:53.4 The entrance pit is used to contain and manage the drilling fluids.

0:57.5 The directional drilling crew are in radio communication with each other throughout the entire process.

VISUAL

Close on the rig front: the drill bit enters the entrance pit and the soil beyond. Drilling fluid collects in the pit (contained). Then a wider view links the rig operator and the locator by radio.

ON-SCREEN TEXT

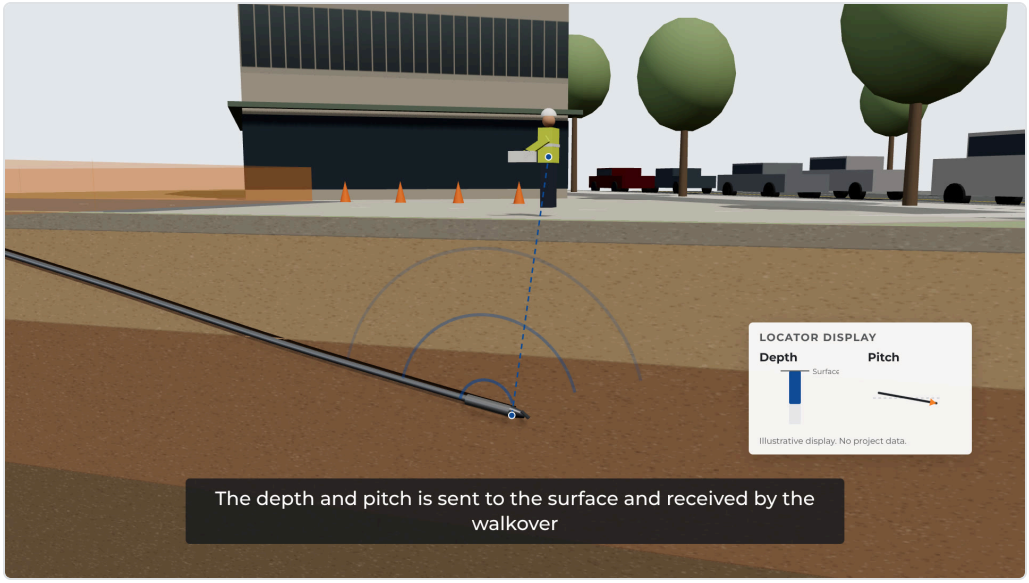
- Entrance pit @ 0:52.0
- Drilling fluid containment @ 0:54.7
- Rig operator @ 0:59.0
- Locator @ 0:59.2
- Radio communication (link between operator and locator)

AUDIENCE FOCUS

Project managers · Environmental project managers · Civil contractors

HDD SPECIALIST REVIEW

Pit position relative to the rig and entry angle; fluid level shown in the pit; radio link graphic.



S07 · 1:04.4 to 1:27.0

S07

NARRATION (ORIGINAL RECORDING)

1:04.4 The directional drill bit is tracked by a sonde, or radio detection device housed within its extended connection to the directional drill pipe.

1:12.9 The depth and pitch is sent to the surface and received by the walkover locating device.

1:18.0 The locator then gathers and interprets the information and gives directions to the directional drill operator on which way to steer the drill bit.

VISUAL

Macro underground: the bit and the sonde housing; the housing turns see-through to show the sonde (transmitter). Signal rings rise to the surface, where the walkover locator receives depth and pitch (illustrative display, no project values). The locator then relays steering directions to the operator.

ON-SCREEN TEXT

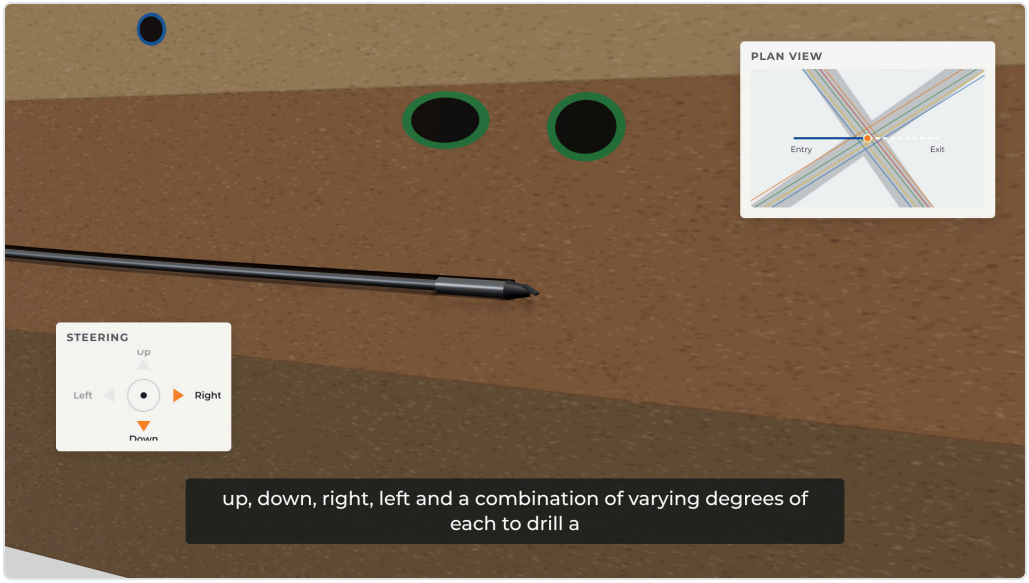
- Sonde (transmitter) @ 1:06.4
- Drill bit @ 1:07.6
- Walkover locator @ 1:15.9
- Locator display: Depth / Pitch (illustrative, no project data)
- Steering directions to the operator (link)

AUDIENCE FOCUS

Civil engineers · Project managers

HDD SPECIALIST REVIEW

Sonde position in its housing behind the bit; locator display layout; signal graphic.



S08 · 1:27.0 to 1:47.7

S08

NARRATION (ORIGINAL RECORDING)

1:27.2 An experienced directional drilling crew can navigate and steer the drill bit up, down, right, left and a combination of varying degrees of each to drill a desired horizontal bore path.

1:38.9 The drilling crew has the ability to determine the exact depth and location of the drill bit at all times.

VISUAL

Follow the drill head along the bore path beneath the intersection. Restrained steering indicators (up, down, right, left) as they are named; a plan inset shows position along the route. Then the view rises to connect the head with the locator walking directly above.

ON-SCREEN TEXT

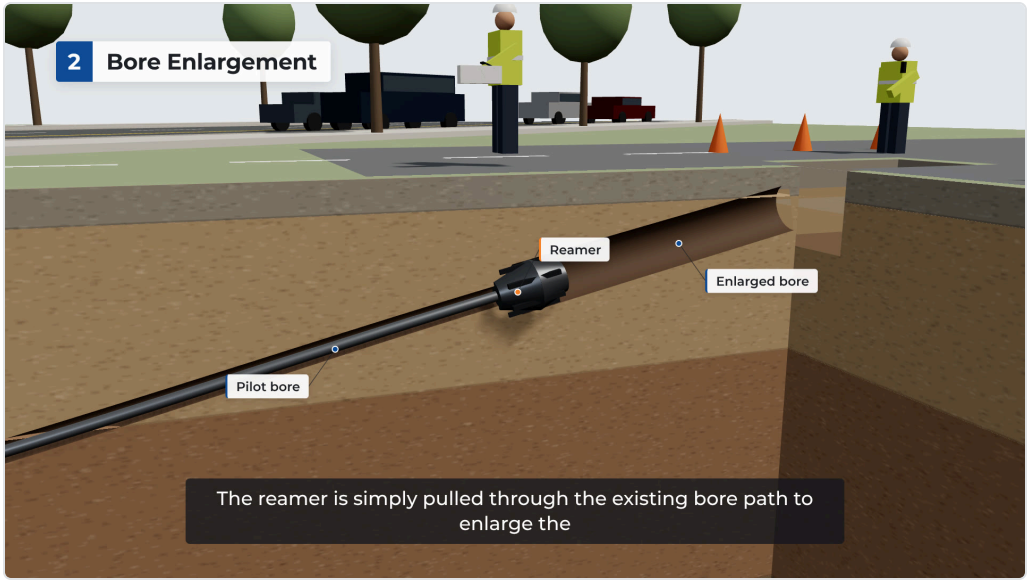
- Tracked position @ 1:41.4
- Steering: Up / Down / Right / Left, then 'Gradual corrections, combined as needed'
- Plan view inset with head position

AUDIENCE FOCUS

Civil engineers · Estimators

HDD SPECIALIST REVIEW

Steering behavior (gradual corrections, no sharp bends) and the tracking line between head and locator.



S09 · 1:47.7 to 2:12.2

S09 · 2. Bore Enlargement

NARRATION (ORIGINAL RECORDING)

1:47.7 A typical mid-sized directional drill uses a 4-inch drill bit to drill a pilot bore.

1:53.4 The bore path is typically enlarged to one and a half size the conduit that is being installed.

1:58.8 In this installation, the bore path has to be enlarged to accommodate the conduit and a reamer is attached to cut a larger diameter hole.

2:07.2 The reamer is simply pulled through the existing bore path to enlarge the hole.

2:12.2 As depicted in this video and as with most directional drilling projects, very little of the normal city activities were interrupted during the conduit installation.

VISUAL

The completed pilot hole in section. A diameter comparison (pilot bore, conduit, enlarged bore) presents the narrated sizes as examples. At the exit pit a reamer is attached to the drill string; it is pulled back toward the rig, cutting the larger hole behind it.

ON-SCREEN TEXT

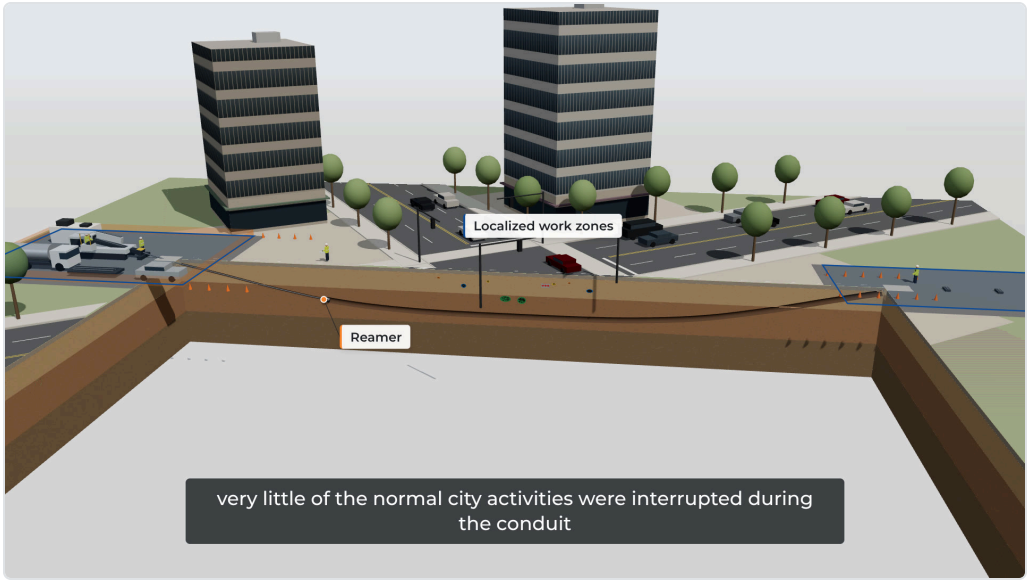
- Pilot bore @ 1:48.6
- Reamer @ 2:03.7
- Enlarged bore @ 2:08.4
- Pilot bore @ 2:08.4
- Sizing (example): pilot bore with 4-inch bit, conduit, enlarged bore about 1.5 × conduit; 'Examples from the narration. Sizes are set for each project.'

AUDIENCE FOCUS

Civil engineers · Estimators · Project managers

HDD SPECIALIST REVIEW

Reamer type and size relative to the pilot hole; direction of travel (pullback toward the rig); example sizing note.



S10 · 2:12.2 to 2:23.9

S10

NARRATION (ORIGINAL RECORDING)

2:12.2 As depicted in this video and as with most directional drilling projects, very little of the normal city activities were interrupted during the conduit installation.

2:23.8 After the bore path is enlarged, the conduit is simply pulled into place by the directional drill rig.

VISUAL

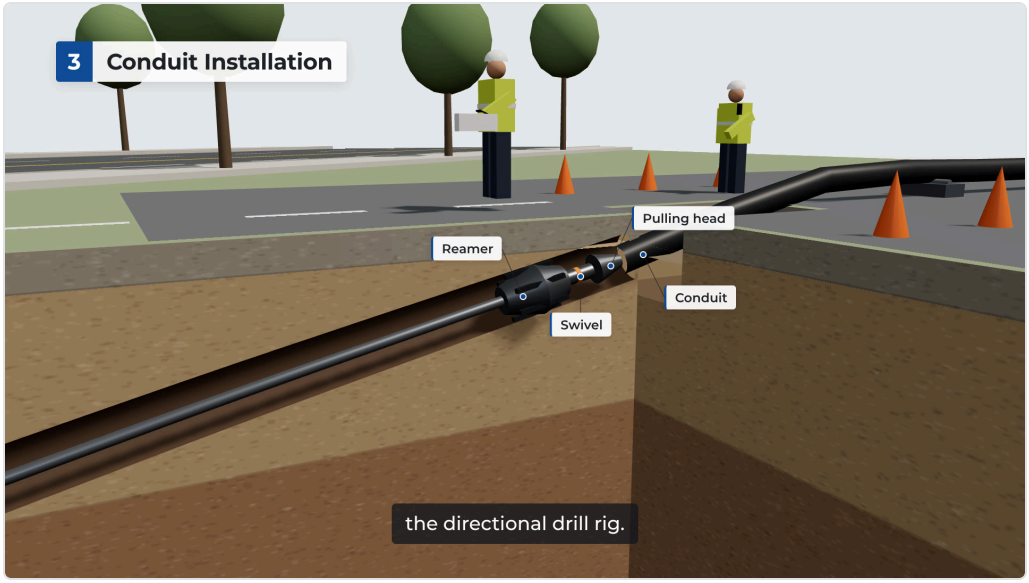
Wide combined view: the cutaway with the reamer moving toward the rig while traffic, signals and pedestrians continue above. Both work zones stay small and visible.

ON-SCREEN TEXT

- Reamer @ 2:13.0
- Localized work zones (outlines)

AUDIENCE FOCUS

Owners and end clients · Civil contractors · Project managers



S11 · 2:23.9 to 2:30.8

S11 · 3. Conduit Installation

NARRATION (ORIGINAL RECORDING)

2:23.8 After the bore path is enlarged, the conduit is simply pulled into place by the directional drill rig.

VISUAL

Exit side: the pullback assembly (reamer, swivel, pulling head) connected to the conduit string laid on rollers. The rig pulls the conduit into the prepared bore toward the rig.

ON-SCREEN TEXT

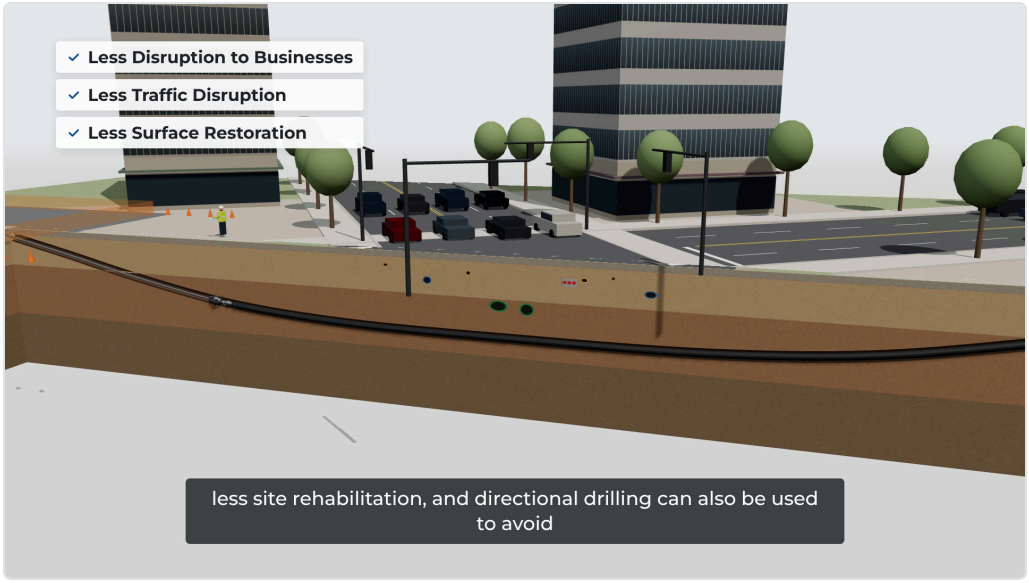
- Reamer @ 2:24.6
- Swivel @ 2:25.0
- Pulling head @ 2:25.4
- Conduit @ 2:26.2

AUDIENCE FOCUS

Project managers · Civil engineers · Civil contractors

HDD SPECIALIST REVIEW

Assembly order (drill pipe, reamer, swivel, pulling head, conduit) and movement direction.



S12 · 2:30.8 to 2:42.9

S12

NARRATION (ORIGINAL RECORDING)

2:30.8 This method of installing utilities has many benefits that include less disruption of commercial activities, less disruption of traffic, less site rehabilitation, and directional drilling can also be used to avoid damage to wetlands, streams, and for river crossings.

VISUAL

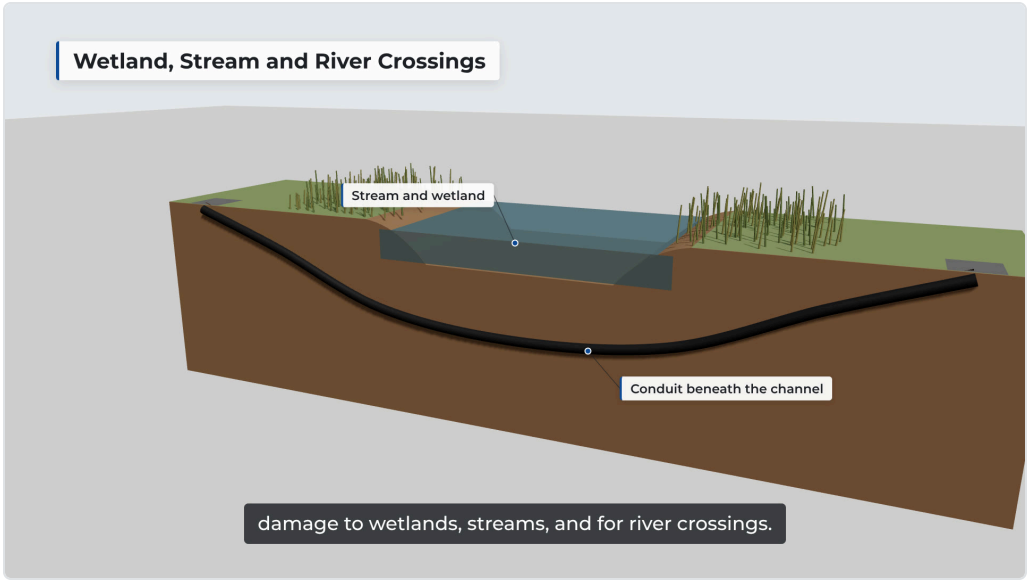
Benefits, as narrated, with short labels: businesses open and accessible, traffic moving through the intersection, pavement left in place while the conduit is pulled underneath (cutaway).

ON-SCREEN TEXT

- Less Disruption to Businesses
- Less Traffic Disruption
- Less Surface Restoration

AUDIENCE FOCUS

Owners and end clients · Project managers · Civil contractors



S13 · 2:42.9 to 2:50.9

S13

NARRATION (ORIGINAL RECORDING)

No narration (silent opening, as in the original).

VISUAL

Separate example site: a stream with wetland banks. A bore passes beneath the channel with a conduit installed, pits set back from the banks. Labeled as a crossing type, with no claim of zero impact.

ON-SCREEN TEXT

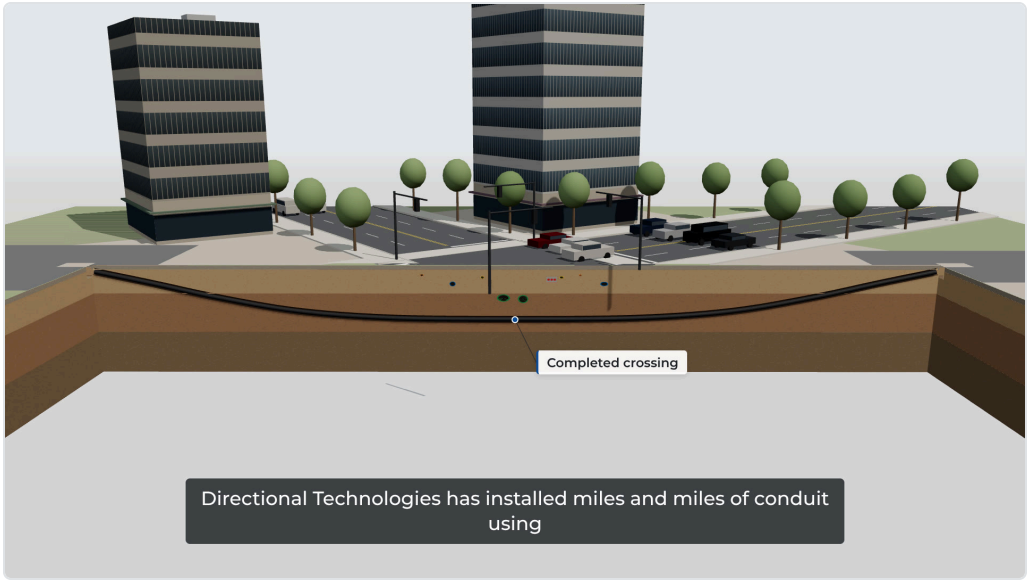
- Wetland, Stream and River Crossings
- Stream and wetland
- Conduit beneath the channel

AUDIENCE FOCUS

Environmental project managers · Civil engineers

HDD SPECIALIST REVIEW

Depth below the channel and setback of the pits from the banks (illustrative).



S14 · 2:50.9 to 2:59.4

S14

NARRATION (ORIGINAL RECORDING)

2:50.9 Directional Technologies has installed miles and miles of conduit using directional drilling **for over 30 years.**

VISUAL

Back at the intersection: the completed crossing. Equipment gone, pits restored, conduit in place beneath the utilities, traffic moving. A short line accompanies 'for over 30 years'.

ON-SCREEN TEXT

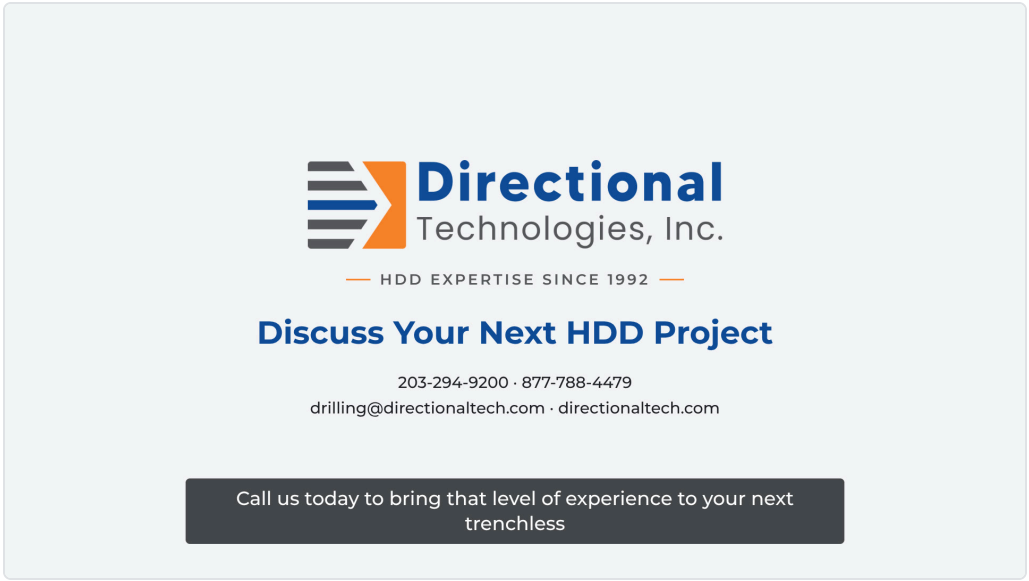
- Completed crossing @ 2:51.6
- Directional Technologies, Inc. / Installing conduit by directional drilling for over 30 years

AUDIENCE FOCUS

Owners and end clients · Project managers

HDD SPECIALIST REVIEW

Final conduit position and restored pits.



S15 · 2:59.4 to 3:08.3

S15

NARRATION (ORIGINAL RECORDING)

2:59.5 Call us today to bring that level of experience to your next trenchless conduit installation.

VISUAL

DTI end card on a neutral background: approved logo, 'HDD Expertise Since 1992', 'Discuss Your Next HDD Project', verified contact details.

ON-SCREEN TEXT

- DTI approved logo
- HDD Expertise Since 1992
- Discuss Your Next HDD Project
- 203-294-9200 · 877-788-4479
- drilling@directionaltech.com · directionaltech.com

AUDIENCE FOCUS

All